

Cloudflare Radar 2024 Year in Review

A data-driven analysis of global Internet traffic, adoption trends, connectivity disruptions, and security threats

Technology Executives & Network Infrastructure Leaders

Source: radar.cloudflare.com/year-in-review/2024

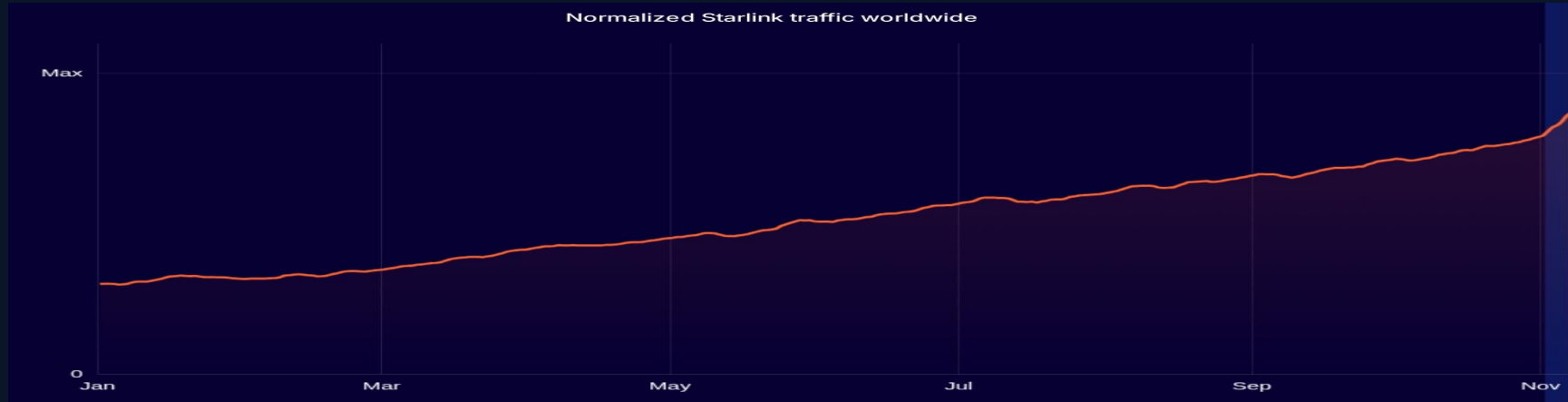
17% Global Traffic Growth and Starlink's 3.3x Surge

17%

Year-over-year growth in global Internet traffic during 2024

3.3x

Starlink traffic surge as satellite connectivity expands to underserved regions



Normalized Starlink traffic worldwide showing 3.3x growth in 2024

Google Leads Top 10 Services as Generative AI Rises

Rank	Service
#1	Google
#2	Facebook
#3	Apple
#4	TikTok
#5	AWS
#6	Microsoft
#7	Instagram
#8	YouTube
#9	Amazon
#10	WhatsApp

Rank	Service
#1	Google
#2	Facebook
#3	Apple
#4	TikTok
#5	AWS
#6	Microsoft
#7	Instagram
#8	YouTube
#9	Amazon
#10	WhatsApp

Key Insight

Generative AI continued its rapid rise throughout 2024, while the Metaverse receded into the background. Rankings based on aggregate data from Cloudflare's 1.1.1.1 DNS resolver.

Post-Quantum Encryption Jumps from 2% to 13% of TLS 1.3



Post-quantum encrypted traffic worldwide showing rapid adoption after Chrome 124 release

2% → 13%

Post-quantum encryption share of TLS 1.3 traffic after Chrome 124 default enablement (April 17)

21%

of traffic uses HTTP/3, running on QUIC for faster, encrypted-by-default connections

Go #1

Go is the top API client language (~12%), followed by Node.js, Python (9.6%), Java (7.4%)

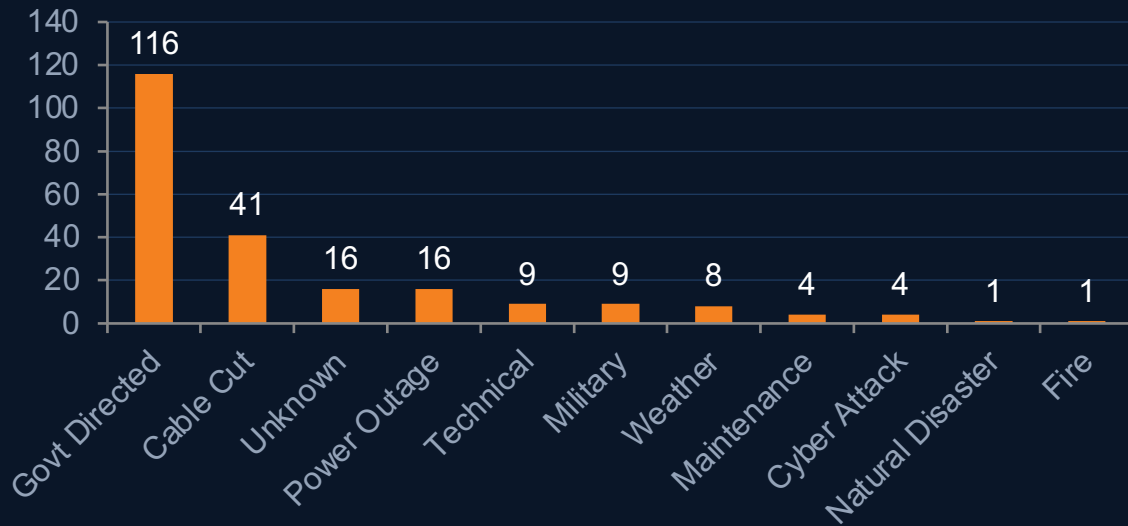
225 Major Disruptions: Government Shutdowns Lead Outage Causes

225

Major Internet disruptions observed globally in 2024

Insight

Government-directed shutdowns (116+) accounted for over half of all major disruptions, underscoring connectivity fragility from policy actions.



6.5% of Traffic Mitigated: Persistent Threats and Bot Concentration

6.5%

of all traffic mitigated by
Cloudflare

Gambling/Games

Most attacked industry sector

68%

of global bot traffic from top 10
countries

Log4j Persistence

CVE-2021-44228, disclosed December 2021, continues to be actively targeted by attackers more than two years later — underscoring the persistent nature of widely known vulnerabilities.

Email Security Threats

4.3% of emails processed were malicious. Of those, 43% contained a deceptive link. The .bar TLD originated the largest share of malicious and spam email observed during the year.

Five Takeaways Shaping the 2025 Internet Outlook

01

Traffic Growth Trajectory

Global Internet traffic grew 17% YoY, with Starlink traffic surging 3.3x as satellite connectivity expanded to underserved regions.

02

Post-Quantum Momentum

Post-quantum encryption adoption leapt from ~2% to 13% of TLS 1.3 traffic following Chrome 124's default enablement in April 2024.

03

Connectivity Fragility

Government-directed shutdowns remained the leading cause of the 225 major Internet disruptions observed, followed by cable cuts (41) and power outages (16).

04

Persistent Security Threats

6.5% of all traffic was mitigated, Gambling/Games was the most targeted sector, and Log4j exploitation persists years after disclosure.

05

Evolving Developer Ecosystem

Go surpassed other languages to become the most popular API client language, capturing roughly 12% of automated API traffic.